

TASK 1: CUSTOMER BEHAVIOUR ANALYSIS

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GITHUB LINK: <https://github.com/Giridharan2007/Ecommerce-Customer-Data-Analysis>

Project Objectives

- Clean and preprocess customer transaction data.
- Analyze customer purchasing behavior.
- Perform customer segmentation using RFM Analysis.
- Identify active and churn-risk customers.
- Visualize purchasing patterns and revenue trends.
- Provide actionable business recommendations.

Dataset Description

The dataset contains customer transaction records including:

- Customer ID
- Product Details
- Purchase Date
- Quantity Purchased
- Product Price
- Order Information

The dataset was obtained from Kaggle and used for customer behavior analysis.

Business Recommendations

Recommendation 1: Loyalty Programs

Introduce reward points and membership benefits for active customers.

Recommendation 2: Personalized Marketing

Recommend products based on previous purchase history.

Recommendation 3: Churn Prevention Campaigns

Send special discounts and offers to inactive customers.

Recommendation 4: Seasonal Promotions

Increase promotional activities during peak sales periods.

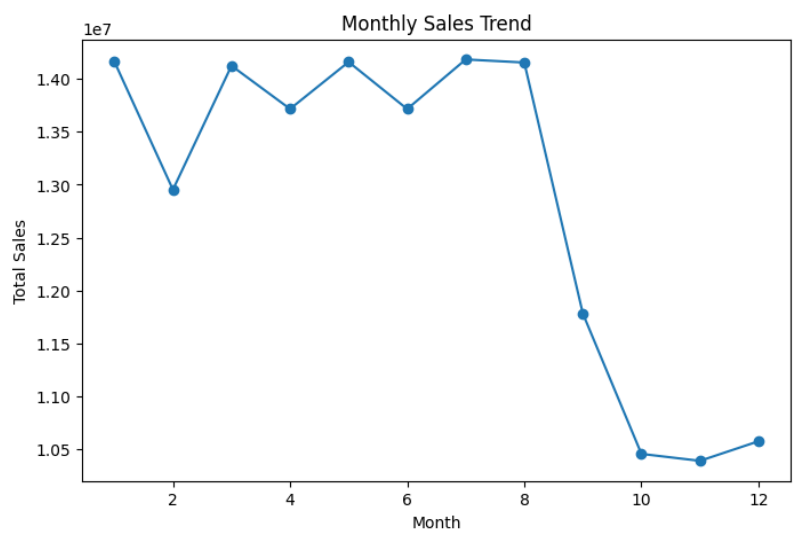
Recommendation 5: Customer Monitoring Dashboard

Track customer engagement and retention metrics regularly.

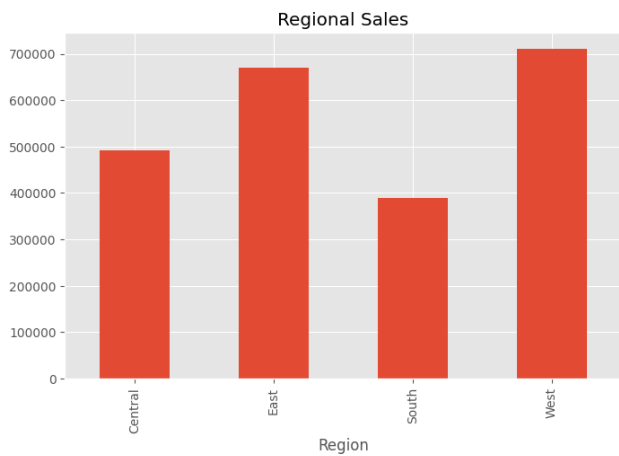
Visual Analysis

The following visualizations were created:

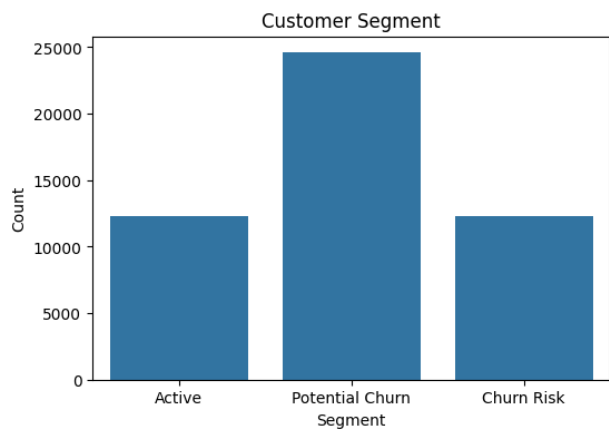
Sales Trend Analysis



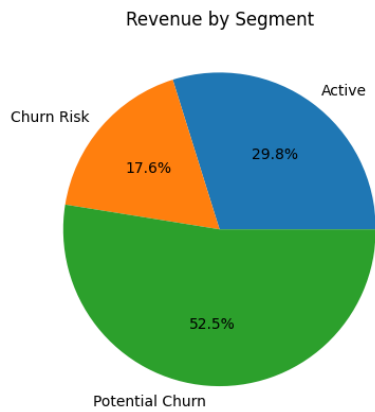
Regional Sales Analysis



Customer Segment Distribution



Revenue Contribution by Segment



Customer Retention Trend



Tools and Technologies Used

- Python
- Google Colab
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-Learn
- Kaggle Dataset

Conclusion

The project successfully analyzed customer purchasing behavior using data analytics techniques. Through data cleaning, feature engineering, RFM segmentation, and visualization, valuable insights were generated regarding customer engagement and spending patterns.

Overall, the project demonstrates how customer behavior analytics can help e-commerce businesses better understand their customers and optimize business performance.